

```
select * from customer limit 50;
```

-- 1. What is the total revenue generated by male vs female?

```
select gender, sum(purchase_amount) as revenue
from customer
group by gender;
```

-- 2. Which customer used discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount from customer
where discount_applied = 'Yes' and
purchase_amount >= (select avg(purchase_amount) from customer)
```

-- 3. Which are the top 5 product with the highest review rating?

```
select item_purchased, round(avg(review_rating::numeric), 2) as review from customer
group by item_purchased
order by avg(review_rating) desc limit 5;
```

-- 4. Compare the average purchase amount between the express shipping and standard shipping.

```
select shipping_type, round(avg(purchase_amount::numeric), 2) as avg_purchase_amt
from customer where shipping_type in ('Standard', 'Express')
group by shipping_type;
```

-- 5. Do subscribed customer spend more? Compare avg. spent and total revenue generated by subscribed and non subscribed customers.

```
select subscription_status, count(customer_id) as total_count, sum(purchase_amount),
round(avg(purchase_amount), 2) as avg_spent, sum(purchase_amount) as total_revenue
from customer
group by subscription_status;
```

-- 6. Which 5 products have the highest % of purchase with discount applied.

```
select item_purchased,
round(100 * sum(case when discount_applied = 'Yes' then 1 else 0 end)/count(*), 2) as
disc_rate
from customer
group by item_purchased
order by disc_rate desc limit 5;
```

-- 7. Segment customers into new, returning and loyal and show the count of each segment.

```
with customer_type as (  
  select customer_id, previous_purchases,  
  case  
    when purchase_frequency_days = 365 then 'New'  
    when purchase_frequency_days between 30 and 90 then 'Returning'  
    else 'Loyal'  
  end as customer_segment  
from customer  
)  
select customer_segment, count(*) as number_of_customers  
from customer_type  
group by customer_segment;
```

--Q8. What are the top 3 most purchased products within each category?

```
WITH item_counts AS (  
  SELECT category,  
    item_purchased,  
    COUNT(customer_id) AS total_orders,  
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)  
DESC) AS item_rank  
  FROM customer  
  GROUP BY category, item_purchased  
)  
SELECT item_rank, category, item_purchased, total_orders  
FROM item_counts  
WHERE item_rank <=3;
```

--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status,  
  COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

--Q10. What is the revenue contribution of each age group?

```
SELECT  
  age_group,  
  SUM(purchase_amount) AS total_revenue  
FROM customer  
GROUP BY age_group  
ORDER BY total_revenue desc;
```

